

Olist SQL Analysis — Insights Summary

Note: This summary is based on the *logic and intent* of each query. Actual figures (dollar amounts, percentages, rankings) will populate once the queries are executed against the live `olist` database. Placeholders below (marked []) should be filled in with real output.

1. Top Customers by Spend (Q1)

Question: Who are the top 10 customers by total amount spent? **Method:** Joins `customers`, `orders`, and `order_payments`; sums `payment_value` per customer, ranks descending.

Insight: A small subset of customers contributes a disproportionate share of revenue — the classic 80/20 pattern common in e-commerce. Once run, the top 10 list identifies who Olist's highest-value customers are, useful for VIP treatment, retention outreach, or loyalty programs.

- Top customer total spend: []
 - Combined spend of top 10: []
 - Share of total revenue from top 10: []%
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2. Monthly Revenue Trend (Q2)

Question: What is the monthly revenue trend across the dataset? **Method:** Aggregates `order_items.price` by purchase month.

Insight: Reveals whether the business is growing, flat, or declining over time, and surfaces seasonality (e.g., holiday spikes). This is the foundation metric for forecasting and goal-setting.

- Peak revenue month: []
 - Lowest revenue month: []
 - Overall trend direction: []
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3. Month-over-Month Revenue Change (Q3)

Question: What is the MoM % change in revenue? **Method:** Uses `LAG()` window function on the monthly revenue series to compute absolute and percentage change.

Insight: Highlights volatility and momentum — sharp month-over-month swings can point to marketing campaigns, supply issues, or seasonal effects worth investigating further.

- Largest MoM increase: []% in []
 - Largest MoM decrease: []% in []
 - Average MoM growth rate: []%
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4. Revenue by Product Category (Q4)

Question: Which product categories generate the most revenue? **Method:** Joins `order_items` and `products`, aggregates revenue, order count, and items sold per category.

Insight: Identifies which categories are the revenue engines of the platform vs. long-tail categories. Useful for inventory prioritization, marketing spend allocation, and category-level growth strategy.

- Top revenue category: []
 - Top 3 categories combined share of revenue: []%
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5. Top 3 Products per Category (Q5)

Question: Which are the top 3 revenue-generating products within each category? **Method:** `ROW_NUMBER()` partitioned by category, ordered by product revenue; unlabeled categories grouped as "Unknown."

Insight: Pinpoints hero products driving each category's performance — useful for merchandising, cross-sell bundles, and identifying which specific SKUs to protect from stockouts.

- Notable standout products: []
 - Categories with a single dominant product (>50% of category revenue): []
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6. Customer Spend Tiers (Q6)

Question: How do customers segment into Low / Medium / High spend tiers? **Method:** CASE logic — Low (<100), Medium (100–500), High (>500) based on total order value.

Insight: Segmentation enables tailored marketing: upsell campaigns for Medium-tier customers, retention/VIP perks for High-tier, and reactivation offers for Low-tier.

- Customers in Low tier: [] ([]%)
 - Customers in Medium tier: [] ([]%)
 - Customers in High tier: [] ([]%)
 - Revenue contribution from High tier: []%
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7. Repeat vs. One-Time Buyers (Q7)

Question: How many customers are repeat buyers vs. one-time buyers? **Method:** Counts distinct orders per customer; classifies as "Repeat buyer" (>1 order) or "One-time buyer" (1 order).

Insight: A low repeat-purchase rate — common in Olist's marketplace data — signals an opportunity to invest in retention (email campaigns, loyalty incentives) rather than relying solely on new customer acquisition.

- One-time buyers: [] ([]%)
 - Repeat buyers: [] ([]%)
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Cross-Cutting Themes

1. **Revenue concentration:** Both customer spend (Q1, Q6) and product performance (Q4, Q5) likely show concentration effects — a minority of customers/products/categories drive the majority of revenue.
2. **Retention over acquisition:** The repeat-buyer analysis (Q7) combined with spend tiers (Q6) suggests retention strategy should be a priority if one-time buyers dominate.
3. **Seasonality & momentum:** The revenue trend and MoM analyses (Q2, Q3) provide the timing context needed to plan promotions around historically strong or weak periods.

Next Steps

- Run all queries against the live `olist` MySQL database and replace placeholders with actual figures.
- Visualize monthly revenue trend and category revenue as charts for the final report/presentation.
- Cross-reference spend tiers with repeat-buyer status to see if High-tier customers are also more likely to be repeat buyers.